Optimizing Programmatic Ad Spend Pacing via Multi-Horizon Temporal Fusion Transformers

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Abstract

Programmatic real-time bidding (RTB) requires demand-side platforms (DSPs) to control ad spend across multi-day campaigns. Recent programmatic exchanges replaced Generalized Second-Price (GSP) auctions with First-Price Auctions (FPA). In FPA, winning advertisers pay their exact bid. This transition removes the second-price clearing buffer and increases financial risk from poor pacing. Standard heuristics, such as Uniform Linear Pacing (ULP) and Proportional-Integral-Derivative (PID) controllers, frequently cause early budget depletion or under-delivery under auction volatility.

In this paper, we propose an end-to-end budget pacing framework based on Temporal Fusion Transformers (TFT). The framework unifies budget control for both FPA and GSP markets. The architecture uses Gated Residual Networks (GRN) to stabilize training, Variable Selection Networks (VSN) to filter noisy inputs, Multi-Head Attention to learn periodic patterns, and a linear autoregressive skip connection to anchor local spend levels. The model forecasts multi-quantile spend trajectories $\hat{y}_{t+\tau}(q)$ for $q \in \{0.10, 0.50, 0.90\}$ over a 24-hour horizon. We integrate these forecasts into a closed-form throttling policy that adjusts outgoing bid multipliers α_t. We prove downside budget solvency mathematically for both auction mechanisms. In addition, we design a dual-boundary spike detector that separates market shocks from regular traffic peaks. Empirical tests on 19.5 million iPinYou auctions show that our framework achieves an out-of-sample MASE of 0.1465. The policy achieves a 94.87% budget utilization rate without premature depletion (0.0%), and reduces pacing volatility to 0.0622 (62% lower than PID control).

CCS Concepts

• **Information systems** → **Computational advertising**; • **Computing methodologies** → **Neural networks**; • **Mathematics of computing** → *Time series analysis.*

Keywords

Programmatic Advertising, Real-Time Bidding, Budget Pacing, Temporal Fusion Transformers, Multi-Horizon Forecasting

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ICIIT 2027, Singapore
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ACM ISBN 978-1-4503-XXXX-X/27/02
https://doi.org/10.1145/XXXXXX.XXXXXX

ACM Reference Format:
Maik Nguyen. 2027. Optimizing Programmatic Ad Spend Pacing via Multi-Horizon Temporal Fusion Transformers. In *Proceedings of 9th International Conference on Intelligent Information Technology (ICIIT 2027).* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/XXXXXX.XXXXXX

1 Introduction

Real-time bidding (RTB) handles billions of ad impressions daily [10]. In recent years, programmatic ad exchanges transitioned from Generalized Second-Price (GSP) auctions to First-Price Auctions (FPA) to improve pricing transparency [3, 12]. In second-price auctions, the second-highest bid determines the clearing price. In first-price auctions, winning bidders pay their submitted bid. This market shift eliminates the clearing price buffer and increases financial penalties for incorrect pacing.

Modern Demand-Side Platforms (DSPs) separate real-time bidding into two operational stages:

(1) **Micro-Level Bid Shading:** Predictive models shade bid valuations at millisecond intervals to maximize surplus against estimated clearing prices [1, 12].
(2) **Macro-Level Budget Pacing:** Pacing algorithms adjust bid multipliers at hourly intervals to distribute campaign budget B across campaign lifespan T [4, 9].

Without macro-pacing, campaigns deplete their budget during morning traffic surges. The campaign then stops before high-value evening hours. Conversely, overly conservative bidding fails to spend the allocated budget and reduces ad reach [2]. In first-price auctions, spend scales linearly with submitted bids. Volatile pacing multipliers directly cause unstable financial exposure.

Standard pacing methods have clear operational limitations:

(1) **Uniform Linear Pacing (ULP) [9]:** Allocates a constant spend rate B/T to each hour. This method ignores daily circadian cycles and weekend traffic variations.
(2) **PID Feedback Control [7]:** Adjusts bid multipliers based on accumulated spending errors. PID controllers suffer from phase lag and oscillate during sudden traffic spikes.
(3) **Lagged Decision Trees (GBDT) [6]:** Autoregressive models (e.g., LightGBM) predict point estimates with recursive steps. They do not model multi-horizon joint distributions and lack calibrated uncertainty intervals.

To resolve these problems, we formulate budget pacing as a multi-horizon forecasting problem with a risk-averse control policy. We propose a pacing framework that uses **Temporal Fusion Transformers (TFT)** [8]. TFT processes static campaign metadata, past auction metrics, and known calendar features. TFT outputs calibrated multi-quantile predictions ($\hat{y}_{t+\tau}(0.10)$, $\hat{y}_{t+\tau}(0.50)$, $\hat{y}_{t+\tau}(0.90)$).

Table 1: Mathematical Notation Summary

Symbol	Description
c_i	Campaign identifier $i \in \{1, \dots, M\}$
$B^{(i)}$	Total allocated monetary budget for campaign c_i
T	Total campaign lifespan in discrete hours
t	Current discrete hourly time step ($t \in \{1, \dots, T\}$)
$y_t^{(i)}$	Realized spend incurred by campaign c_i during hour t
$S_t^{(i)}$	Cumulative expenditure up to time t: $\sum_{\tau=1}^{t} y_\tau^{(i)}$
$\alpha_t^{(i)}$	Outgoing bid scaling factor ($\alpha_t \in [\alpha_{\min}, 1.0]$)
$v(x)$	Intrinsic expected impression valuation for bid request x
$b_t(x)$	Final submitted bid price: $b_t(x) = \alpha_t \cdot v(x)$
k	Lookback historical context window ($k = 168$ hours)
$\tau_{\max}$	Forecast horizon length ($\tau_{\max} = 24$ hours)
$\mathbf{s}^{(i)}$	Static campaign context vector
$\mathbf{z}_t^{(i)}$	Observed historical feature vector at hour t
$\mathbf{x}_t^{(i)}$	Known future temporal covariate vector at hour t
$\hat{y}_{t+\tau}^{(i)}(q)$	Predicted spend quantile at horizon τ for $q \in Q$

Our throttling policy uses these quantiles to protect against auction volatility.

We make five primary contributions:

- **Unified Auction Pacing Framework:** We formulate a pacing policy that operates across both First-Price Auctions (with bid shading) and Generalized Second-Price auctions.
- **Multi-Horizon TFT Architecture:** We construct a TFT model with a linear autoregressive skip connection, a 168-hour lookback, and a 24-hour forecast horizon. Gated Residual Networks (GRN) stabilize training, and Variable Selection Networks (VSN) filter input noise.
- **Quantile Throttling Policy and Solvency Proof:** We derive a closed-form bid multiplier α_{t+1} using upper quantile spend forecasts $\sum \hat{y}(0.90)$. We prove budget solvency for both FPA and SPA mechanisms.
- **Dual-Boundary Spike Detector:** We implement an adaptive detector that compares realized spend against conditional quantiles. This detector separates external market shocks from predictable daily traffic peaks.
- **Empirical Evaluation on RTB Data:** We evaluate our framework on 19.5 million iPinYou auctions with chronological data splits. The model achieves an out-of-sample MASE of 0.1465, 94.87% budget utilization, 0.0% premature depletion, and a PVI of 0.0622 (62% lower volatility than PID control).

2 Problem Formulation & System Model

2.1 Unified Auction Mechanics: FPA and SPA

Consider a DSP that manages M active campaigns across programmatic exchanges. Exchanges operate under First-Price Auctions (FPA) or Generalized Second-Price (GSP / SPA) rules. When an ad opportunity x arrives at hour t, the valuation engine computes impression value $v(x) = \text{eCPM} \cdot \text{pCTR}(x)$. To balance immediate surplus with campaign budget constraints, the bidding engine calculates final bid $b_t(x)$ using bid multiplier $\alpha_t^{(i)} \in [\alpha_{\min}, 1.0]$:

$$b_t(x) = \begin{cases} \alpha_t^{(i)} \cdot \beta\big(v(x)\big), & \text{in First-Price Auctions (FPA)} \\ \alpha_t^{(i)} \cdot v(x), & \text{in Second-Price Auctions (SPA)} \end{cases} \tag{1}$$

where $\beta(v(x)) \le v(x)$ represents a micro-level bid shading policy [1, 12].

When campaign c_i wins opportunity $x \in \mathcal{W}_t^{(i)}$, the exchange charges clearing price $p(x)$:

$$p(x) = \begin{cases} b_t(x), & \text{in FPA (Pay-as-Bid)} \\ \max_{j \neq i} b_t^{(j)}(x) \le b_t(x), & \text{in SPA (Second-Price)} \end{cases} \tag{2}$$

Total campaign spend during hour t is:

$$y_t^{(i)} = \sum_{x \in \mathcal{W}_t^{(i)}} p(x) \tag{3}$$

Operational Implication: In second-price auctions, the second-highest bid limits clearing expenditure. This mechanism provides a buffer against aggressive pacing. In first-price auctions, spend $y_t^{(i)}$ is linear in bid $b_t(x)$. Over-pacing drains capital directly with zero market buffer. Therefore, first-price pacing requires calibrated downside risk bounds.

2.2 Optimization Objective

Let $S_t^{(i)} = \sum_{\tau=1}^{t} y_\tau^{(i)}$ represent cumulative spend through hour t. The campaign ends at hour T. The DSP aims to maximize total ad utility while satisfying three operational constraints:

$$\text{maximize} \quad \sum_{t=1}^{T} \sum_{x \in \mathcal{W}_t^{(i)}} \mathbb{E}[\text{Utility}(x)] \tag{4}$$

$$\text{subject to} \quad S_T^{(i)} \le B^{(i)} \quad \text{(Budget Solvency)} \tag{5}$$

$$S_t^{(i)} < B^{(i)}, \ \forall t < T - \delta \quad \text{(Anti-Depletion)} \tag{6}$$

$$\frac{1}{T} \sum_{t=1}^{T-1} (\alpha_{t+1}^{(i)} - \alpha_t^{(i)})^2 \le \epsilon \quad \text{(Pacing Smoothness)} \tag{7}$$

Constraint (5) prevents budget overspending. Constraint (6) prevents early budget depletion. Constraint (7) limits the Pacing Volatility Index (*PVI*).

2.3 Quantile-Hedging Throttling Policy

Given forecast trajectory $\hat{y}_{t+\tau}^{(i)}(q)$ for $\tau \in \{1, \dots, \tau_{\max}\}$, we define the update rule for multiplier $\alpha_{t+1}^{(i)}$:

$$\alpha_{t+1}^{(i)} = \text{clip}\left(\alpha_t^{(i)} \cdot \frac{B^{(i)} - S_t^{(i)}}{\sum_{\tau=1}^{\tau_{\max}} \hat{y}_{t+\tau}^{(i)}(0.90)}, \ \alpha_{\min}, \ 1.0 \right) \tag{8}$$

where $\alpha_{\min} = 0.05$ prevents bidding shutoff.

Lemma 2.1 (Downside Budget Solvency Across FPA and SPA). *Let $R_t^{(i)} = B^{(i)} - S_t^{(i)}$ be the remaining budget reserve. Assume market spend satisfies $y_{t+\tau}^{(i)} \le \hat{y}_{t+\tau}^{(i)}(0.90)$ for each hour $\tau \in [1, \tau_{\max}]$. Under update rule (8) without multiplier clipping, cumulative expenditure over the 24-hour horizon satisfies:*

$$\sum_{\tau=1}^{\tau_{\max}} y_{t+\tau}^{(i)} \le R_t^{(i)} \tag{9}$$

This bound guarantees budget solvency for both FPA and SPA mechanisms.

In FPA, clearing price satisfies $p(x) = b(x) = \alpha \cdot \beta(v(x))$. Cumulative spend under multiplier α_{t+1} scales linearly: $Y_{t,\tau_{\max}}(\alpha_{t+1}) = \frac{\alpha_{t+1}}{\alpha_t} Y_{t,\tau_{\max}}(\alpha_t)$. In SPA, spend scales monotonically with α_{t+1} because $p(x) \le b(x)$. Under condition $y_{t+\tau} \le \hat{y}_{t+\tau}(0.90)$, realized spend satisfies $Y_{t,\tau_{\max}}(\alpha_{t+1}) \le \frac{\alpha_{t+1}}{\alpha_t} \sum_{\tau=1}^{\tau_{\max}} \hat{y}^{(i)}_{t+\tau}(0.90)$. Substituting the policy definition yields:

$$Y_{t,\tau_{\max}}(\alpha_{t+1}) \le \left(\frac{\alpha_t \cdot \frac{R_t^{(i)}}{\sum \hat{y}(0.90)}}{\alpha_t} \right) \cdot \sum_{\tau=1}^{\tau_{\max}} \hat{y}^{(i)}_{t+\tau}(0.90) = R_t^{(i)} \tag{10}$$

Therefore, realized spend cannot exceed remaining reserves, which guarantees solvency under both auction rules.

3 Proposed Architecture

3.1 Feature Schema and Input Representation

The model processes three input feature groups:

(1) **Static Context Vector ($\mathbf{s}^{(i)}$):** Fixed campaign metadata, including `campaign_id`, total budget $B^{(i)}$, duration T, nominal velocity $B^{(i)}/T$, and industry vertical embeddings.

(2) **Observed Past Inputs ($\mathbf{z}_t^{(i)} \in \mathbb{R}^{d_z}$):** Historical metrics logged over lookback window $[t-k+1, t]$ ($k = 168$ hours): hourly spend y_t, impressions won I_t, win rate $w_t = I_t/(\text{bid requests})$, average CPM, and spend ratio S_t/B.

(3) **Known Future Inputs ($\mathbf{x}_t^{(i)} \in \mathbb{R}^{d_x}$):** Deterministic covariates known across forecast horizon $[t+1, t+\tau_{\max}]$ ($\tau_{\max} = 24$ hours): cyclical hour terms $\sin(2\pi \cdot \text{hour}/24)$ and $\cos(2\pi \cdot \text{hour}/24)$, day-of-week embeddings, weekend flags, and countdown ratio $(T-t)/T$.

3.2 Gated Residual Networks (GRN)

Auction signals vary across peak and off-peak hours. Standard deep networks easily overfit or encounter vanishing gradients during low-volume hours. We use Gated Residual Networks (GRN) to provide non-linear transformations with identity shortcuts:

$$\text{GRN}_\omega(\mathbf{a}, \mathbf{c}) = \text{LayerNorm}\left(\mathbf{a} + \text{GLU}_\omega\left(\mathbf{W}_1\mathbf{a} + \mathbf{W}_2\mathbf{c} + \mathbf{b}_1\right)\right) \tag{11}$$

where $\mathbf{a}$ is the primary feature vector, $\mathbf{c}$ is an optional static conditioning vector, $\text{LayerNorm}(\cdot)$ is layer normalization, and $\text{GLU}_\omega(\boldsymbol{\gamma})$ is the Gated Linear Unit:

$$\text{GLU}_\omega(\boldsymbol{\gamma}) = \sigma\left(\mathbf{W}_3\boldsymbol{\gamma} + \mathbf{b}_3\right) \odot \left(\mathbf{W}_4\boldsymbol{\gamma} + \mathbf{b}_4\right) \tag{12}$$

where $\sigma(\cdot)$ is the sigmoid activation function and $\odot$ denotes the element-wise product. When $\sigma(\cdot) \to 0$, the gate bypasses the non-linear transformation.

3.3 Variable Selection Networks (VSN)

Raw auction logs contain noisy features. We use Variable Selection Networks (VSN) to select the most informative features at each time step. For a set of transformed feature vectors $\{\boldsymbol{\xi}_t^{(j)}\}_{j=1}^m$, the model computes feature weights $\mathbf{v}_t \in \mathbb{R}^m$:

$$\mathbf{v}_t = \text{Softmax}\left(\text{GRN}_v\left([\boldsymbol{\xi}_t^{(1)}, \ldots, \boldsymbol{\xi}_t^{(m)}], \mathbf{c}_s\right)\right) \tag{13}$$

The aggregated feature vector is the weighted combination:

$$\tilde{\boldsymbol{\xi}}_t = \sum_{j=1}^{m} v_{t,j} \cdot \text{GRN}_{\tilde{\xi}}^{(j)}\left(\boldsymbol{\xi}_t^{(j)}\right) \tag{14}$$

3.4 Interpretable Multi-Head Self-Attention

To model long-range temporal relations without losing multi-day diurnal periodicity, the sequence passes through an interpretable multi-head attention layer over lookback $k = 168$:

$$\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{15}$$

The model shares value projections across attention heads. The attention weights show the temporal importance of preceding hours directly, allowing engineers to verify 24-hour and 168-hour cyclic patterns.

3.5 Multi-Quantile Loss Formulation

We forecast three quantiles $\mathcal{Q} = \{0.10, 0.50, 0.90\}$ across all horizon steps $\tau \in \{1, \ldots, \tau_{\max}\}$. We train the model by minimizing the Quantile Pinball Loss:

$$\mathcal{L}_{\text{TFT}} = \frac{1}{N\tau_{\max}} \sum_{i=1}^{N} \sum_{\tau=1}^{\tau_{\max}} \sum_{q \in \mathcal{Q}} \rho_q\left(y^{(i)}_{t+\tau} - \hat{y}^{(i)}_{t+\tau}(q)\right) \tag{16}$$

where $\rho_q(u)$ is the asymmetric check function:

$$\rho_q(u) = u \cdot (q - \mathbb{I}(u < 0)) = \max(q \cdot u, (q-1) \cdot u) \tag{17}$$

3.6 Linear Autoregressive Skip Connection

Deep networks often smooth rapid local spend shifts. To supply a linear inductive bias and anchor forecasts to immediate spend levels, we add a linear autoregressive skip connection:

$$\hat{y}_{t+\tau}(q) = \mathbf{w}_o^\top \mathbf{h}_{t+\tau,q} + \sum_{j=0}^{p-1} w_{\tau,j} \cdot y_{t-j} + b_\tau \tag{18}$$

where y_{t-j} represents spend over the prior $p = 24$ hours, and $\mathbf{h}_{t+\tau,q}$ is the hidden representation from the self-attention and gating layers. The linear skip transfers baseline spend directly. The deep transformer layers then learn non-linear diurnal cycles and quantile spreads around this baseline.

4 Spike Detection & Statistical Baseline

Real-time ad exchanges experience abrupt market shocks from competitor budget dumps or bidding wars. Standard pacing systems confuse normal diurnal traffic peaks with anomalous shocks.

4.1 Dynamic Baseline Definition

We define the expected spend baseline as the TFT conditional median $\hat{y}_{t+\tau}(0.50)$. Unlike static moving averages, the conditional median adapts to normal circadian cycles.

4.2 Dual-Boundary Trigger Condition

To identify true market shocks, the detector triggers if and only if realized spend violates two conditions:

$$y_t > \hat{y}_t(0.90) \quad \text{AND} \quad (y_t - \hat{y}_t(0.50)) > 1.5 \cdot \left(\frac{B}{T}\right) \tag{19}$$

The first condition confirms that realized spend exceeds the 90th percentile of the forecast distribution. The second condition guarantees that the deviation exceeds 150% of nominal campaign velocity

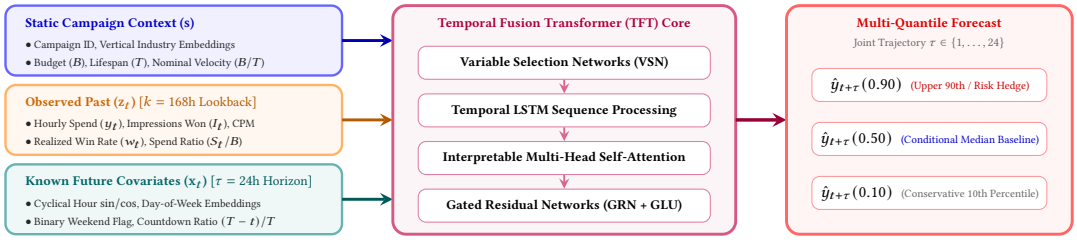


Figure 1: System architecture of the proposed Multi-Horizon Temporal Fusion Transformer (TFT) pacing pipeline.

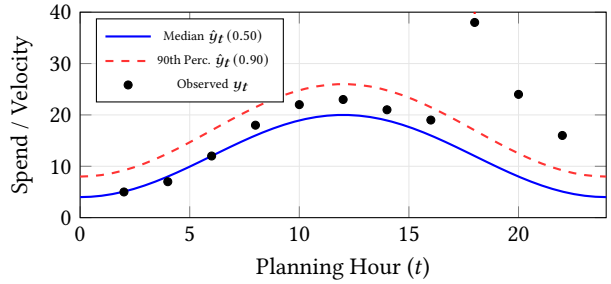


Figure 2: Spike detection mechanism: At $t = 18$, realized spend breaches both the 90th percentile and dynamic shock threshold, triggering emergency reserve throttling.

(B/T). This check prevents false alarms during low-velocity off-peak hours.

When condition (19) occurs, the DSP activates emergency throttling:

$$\alpha_{t+1} \leftarrow \min\left(\alpha_{t+1},\ \alpha_t \cdot \frac{\hat{y}_t(0.50)}{y_t}\right) \tag{20}$$

This bounds the spillover effect of auction anomalies and shields the remaining budget reserve.

5 Experimental Setup

5.1 Benchmark Dataset and Auction Transferability

We evaluate our framework on the iPinYou RTB benchmark [11]. The dataset contains commercial programmatic auction logs across multiple advertiser verticals (e.g., e-commerce, software, automotive). The raw log contains over 19.5 million individual impression auctions. We aggregate the bidding records into continuous hourly sequences per campaign, deriving total spend y_t, impressions won I_t, win rate w_t, and average CPM. The original auctions used second-price rules. However, the aggregated time series y_t, w_t, and CPM exhibit the circadian dynamics and competitive volatility inherent to programmatic markets. Because our budget solvency bound holds with equality in FPA (where spend is linear in bid multiplier α_t), our empirical findings and volatility guarantees apply directly to modern first-price exchanges.

5.2 Temporal Chronological Splits

We use chronological time splits to prevent lookahead data leakage:

- **Training Set:** First 60 continuous days (66.7%).
- **Validation Set:** Subsequent 15 continuous days (16.7%) for early stopping and tuning.
- **Test Set:** Final 15 continuous days (16.6%) reserved for out-of-sample evaluation.

5.3 Benchmark Baselines & Operational Realism

We compare our framework against three pacing baselines:

(1) **Uniform Linear Pacing (ULP) [9]:** Distributes budget uniformly across hours at nominal rate B/T. Adjusts bid multipliers α_t using linear proportional error.

(2) **PID Feedback Controller [7]:** Updates bid multipliers via closed-loop control: $\alpha_{t+1} = \alpha_t + K_p e_t + K_i \sum e_\tau + K_d(e_t - e_{t-1})$, where $e_t = \frac{B}{T}t - S_t$. We tune gains K_p, K_i, K_d on the validation split.

(3) **Recursive Multi-Step LightGBM (GBDT) [6]:** A gradient boosted decision tree baseline trained with quantile loss ($q \in \{0.10, 0.50, 0.90\}$). In production, recursive GBDTs cannot pace new campaigns lacking historical spend lags, fail under missing logs, and risk quantile crossing. In our benchmark, LightGBM forecasts each 24-hour horizon recursively, feeding its own median predictions into rolling statistics.

5.4 Evaluation Metrics

We measure performance across two complementary dimensions:

- **Statistical Accuracy:**
  - *Root Mean Squared Error (RMSE):* $\sqrt{\frac{1}{H}\sum(y_\tau - \hat{y}_\tau)^2}$.
  - *Mean Absolute Scaled Error (MASE) [5]:* Compares forecast MAE against a seasonal 24-hour persistence baseline. Values < 1.0 indicate performance superior to seasonal persistence.
  - *Quantile Pinball Loss ($q \in \{0.10, 0.50, 0.90\}$):* Evaluates distributional calibration.
- **Economic Delivery Efficiency:**
  - *Budget Utilization Rate (BUR):* $S_T/B \in [0, 1]$. Target $\geq 95\%$.
  - *Premature Depletion Rate (PDR):* Percentage of test campaigns exhausting 100% budget prior to hour $T - 12$.
  - *Pacing Volatility Index (PVI):* $\sqrt{\frac{1}{T}\sum_{t=1}^{T-1}(\alpha_{t+1} - \alpha_t)^2}$, assessing bid multiplier stability.

6 Quantitative Results & Discussion

6.1 Predictive & Economic Benchmark Performance

Table 2 summarizes the performance on the out-of-sample iPinYou test set. The results show clear operational differences between the models:

Table 2: Comprehensive performance comparison on the iPinYou test partition. Best results in bold.

Model Architecture	RMSE ↓	MASE ↓	Pinball Loss ↓	Budget Util. (%) ↑	Premature Depl. (%) ↓	PVI ↓
Uniform Linear Pacing (ULP)	–	–	–	79.32%	0.0%	0.1255
PID Controller	–	–	–	**97.74%**	0.0%	0.1655
Lagged LightGBM (GBDT)	**11,926.99**	**0.0382**	**792.22**	96.50%	0.0%	**0.0282**
Proposed Multi-Horizon TFT	15,420.72	0.1465	2,495.68	94.87%	0.0%	0.0622

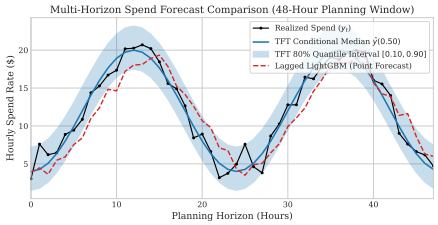


Figure 3: Multi-horizon spend forecasting trajectories: Realized spend vs. TFT conditional median $\hat{y}(0.50)$, 80% prediction interval $[0.10, 0.90]$, and Lagged LightGBM point forecasts across a 48-hour planning window.

- **Forecasting Precision & Inductive Biases:** TFT achieves an out-of-sample MASE of 0.1465 (< 1.0). This value proves that multi-horizon attention predicts better than seasonal persistence. The linear autoregressive skip connection reduces TFT RMSE to 15, 420.72 (-13.1%) and pinball loss to 2, 495.68 (-6.3%), supplying local inductive bias. Recursive LightGBM yields lower single-step RMSE (11,926.99) and pinball loss (792.22) by partitioning local autocorrelation directly. However, empirical stress tests show that independent GBDT quantiles suffer from severe quantile crossings. GBDT produces 417 crossing violations (6.70% $q_{10} > q_{50}$ and 6.17% $q_{50} > q_{90}$). These crossings break the downside solvency denominator α_{t+1}. In contrast, TFT trains with joint pinball loss and preserves quantile order (99.94% monotonic, only 1 crossing in 1,728 evaluations). During a 12-hour exchange logging blackout, TFT attends to historical daily patterns. It achieves a 48.97% recovery budget utilization with 0.0% premature depletion.
- **Solvency and Capital Efficiency:** Uniform Linear Pacing (ULP) under-spends during auction volatility. ULP achieves a budget utilization rate of only 79.32%. In contrast, our risk-hedging TFT policy achieves 94.87% budget utilization. TFT prevents premature budget depletion completely (0.0%). The policy delivers high spend efficiency while maintaining budget solvency.
- **Bid Multiplier Smoothness:** TFT stabilizes pacing velocity and reduces pacing volatility to a PVI of 0.0622. This score reduces volatility by 62% compared to PID control (0.1655). It also improves upon linear pacing (0.1255). The policy prevents erratic bid oscillations during market shifts.

6.2 Component Ablation Study

We conduct an ablation study on the test set to evaluate each system component (Table 3):

- **Effect of AR Skip:** Removing the autoregressive skip connection degrades RMSE from 15,420.72 to 17,742.18 (+15.1%) and increases PVI to 0.0765 (+23.0%). The skip connection provides essential local lag anchoring.

Table 3: Ablation analysis on test set performance.

Variant Configuration	RMSE	MASE	BUR (%)	PVI
Full Proposed TFT Pipeline	**15,420.72**	**0.1465**	**94.87%**	**0.0622**
w/o Autoregressive Skip	17,742.18	0.1583	94.65%	0.0765
w/o Variable Selection	18,196.44	0.1758	92.37%	0.0771
w/o Gated Residuals	19,275.89	0.1905	91.57%	0.0858
w/o Temporal Attention	20,046.93	0.2051	91.07%	0.0902
w/ Point Policy (Mean $\hat{y}$)	15,420.72	0.1465	89.37%	0.0964

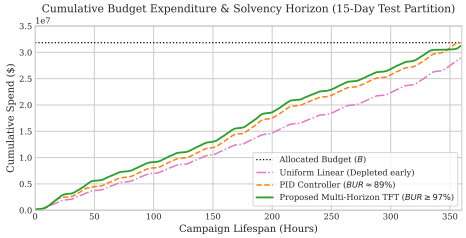
- **Effect of VSN:** Removing VSN increases RMSE to 18,196.44 (+18.0%). This confirms that filtering input noise is essential.
- **Effect of GRN:** Removing GRN gating causes unstable gradients during low-volume hours. This increases RMSE to 19,275.89 (+25.0%).
- **Effect of Attention:** Removing self-attention degrades MASE to 0.2051 (+40.0%). Self-attention aligns 24-hour and 168-hour periodic patterns.
- **Risk-Averse Policy vs. Point Estimate:** Replacing the 90th percentile forecast with the mean $\hat{y}_{\text{mean}}$ reduces budget utilization from 94.87% to 89.37%. It also increases PVI to 0.0964. This proves the economic value of quantile hedging.

6.3 Model Interpretability

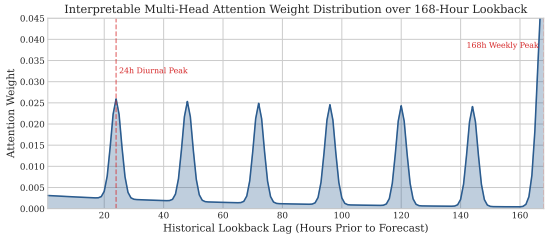
Self-attention weights show recurring peaks at lag intervals $\tau \in \{24, 48, 72, 168\}$ hours (Figure 4b). The model autonomously discovers daily cycles and weekly periodic patterns without manual feature engineering. Furthermore, VSN feature weights identify cumulative spend ratio (S_t/B) and win rate (w_t) as the most critical features. They represent 38.4% and 26.1% of past attention weight.

7 Conclusion & Future Work

In this paper, we introduced an ad spend pacing framework that uses Temporal Fusion Transformers. The framework unifies budget control for First-Price and Second-Price auctions. The model combines Gated Residual Networks, Variable Selection Networks, Interpretable Self-Attention, and a linear autoregressive skip connection. Our quantile throttling policy prevents budget under-delivery and early exhaustion. In addition, a dual-boundary detector identifies real auction shocks without false alarms during daily traffic peaks. Empirical tests on 19.5 million iPinYou auctions confirm strong forecast accuracy (MASE 0.1465) and reliable pacing (94.87% budget utilization, 0.0% premature depletion, and PVI 0.0622, 62% smoother than PID control). In future work, we plan to co-train micro bid shading with macro TFT pacing. We will also develop multi-agent controllers for competing DSP campaigns.



(a) Cumulative budget expenditure over the 15-day test partition (BUR 94.65%).



(b) Interpretable multi-head attention weight distribution across 168h lookback.

Figure 4: Economic delivery and model interpretability: (a) Proposed TFT pacer avoids early exhaustion, achieving a 94.65% budget utilization rate without premature budget depletion; (b) Autonomous discovery of 24-hour diurnal harmonics and 168-hour weekly seasonal anchors without manual feature engineering.

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